STPM PHYSICS · SEMESTER 1 · CHAPTER 5

Circular Motion

5.1 Angular Velocity and Centripetal Acceleration

DEFINITIONS & LAWS

- **Angular displacement** θ is the angle swept out by the radius vector from the centre of the circle, measured in radian. 1 rad is the angle subtended at the centre of a circle by an arc equal in length to the radius.
- **Angular velocity** ω is the rate of change of angular displacement, $\omega = \frac{d\theta}{dt}$ (rad s^{-1}).
- **Period** T is the time taken for one complete revolution; **frequency** f is the number of revolutions per unit time, $f = \frac{1}{T}$ (Hz).
- **Uniform circular motion** is motion in a circle at constant speed. The velocity changes continuously in direction, so the object accelerates.
- **Centripetal acceleration** is the acceleration of an object in circular motion; it is directed towards the centre of the circle, perpendicular to the velocity, and changes the direction of the velocity but not the speed.
- Every point on a rigid rotating body has the same ω and T, but the linear speed v increases with distance r from the axis.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Angular displacement	$\theta = \frac{s}{r}$	θ = angle (rad); s = arc length (m); r = radius (m); 360° = 2π rad
Angular velocity	$\omega = \frac{d\theta}{dt} = \frac{2\pi}{T} = 2\pi f$	ω = angular velocity (rad s^{-1}); T = period (s); f = frequency (Hz)
Linear speed in circular motion	$v = r\omega$	v = linear (tangential) speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
Centripetal acceleration	$a = \frac{v^2}{r} = r\omega^2 = v\omega$	a = centripetal acceleration (m s^{-2}), directed towards the centre; v = speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
Angular velocity (constant α)	$\omega = \omega_0 + \alpha t$	ω_0 = initial angular velocity; ω = final angular velocity (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); t = time (s)
Angular displacement (constant α)	$\theta = \omega_0 t + \frac{1}{2}\alpha t^2$	θ = angular displacement (rad); ω_0 = initial angular velocity (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); t = time (s)

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Angular velocity–displacement (constant α)	$\omega^2 = \omega_0^2 + 2\alpha\theta$	ω_0, ω = initial and final angular velocities (rad s^{-1}); α = constant angular acceleration (rad s^{-2}); θ = angular displacement (rad)
Angular displacement from mean ω (constant α)	$\theta = \frac{1}{2}(\omega_0 + \omega)t$	θ = angular displacement (rad); ω_0, ω = initial and final angular velocities (rad s^{-1}); t = time (s); rotation with constant angular acceleration

HOW TO ANSWER (MPM STYLE)

- For a definition, state the quantity, "rate of change of angular displacement", and the unit rad s^{-1}.
- Convert first: rev min^{-1} → rad s^{-1} by × $\frac{2\pi}{60}$; degrees → rad by × $\frac{\pi}{180}$; cm → m.
- Write formula → substitute with units → answer to 3 s.f. with unit; state the direction of a as "towards the centre".
- To explain why a is non-zero at constant speed: the direction of the velocity changes, so the velocity changes, hence acceleration exists.

SECTION A TRAPS

Trap: Constant speed does not mean constant velocity or zero acceleration; the acceleration is $\frac{v^2}{r}$ towards the centre.

Trap: Using rev min^{-1} or degrees directly in $v = r\omega$ or $a = r\omega^2$; ω must be in rad s^{-1}. Also using diameter in place of radius.

Trap: Two points on the same rotating disc have equal ω and T but different v and a; the point further from the axis has the larger $v = r\omega$ and $a = r\omega^2$.

TRICKS

Trick: Ratio method: at fixed ω, $v \propto r$ and $a \propto r$; at fixed v, $a \propto \frac{1}{r}$. Doubling ω at fixed r gives 4 times the acceleration.

Trick: Unit check: $\frac{v^2}{r}$ gives m s^{-2}, and $r\omega^2$ gives m s^{-2} only when ω is in rad s^{-1} (s^{-1}).

5.2 Centripetal Force

DEFINITIONS & LAWS

- **Centripetal force** is the resultant force directed towards the centre of the circle that is required to make an object move in a circular path; it is perpendicular to the velocity and does no work.
- Centripetal force is not a separate force; it is provided by tension, friction, gravitational force, normal reaction or a component of these forces.

- **Newton's first law**: a body remains at rest or moves with constant velocity unless acted on by a resultant force; hence a body in circular motion needs a resultant force towards the centre.
- Explain the "outward" sensation: it is due to inertia of the body, which tends to continue in a straight line (tangent); there is no real outward force on the body in an inertial frame.
- If the centripetal force is removed (string breaks), the object moves along the tangent at constant velocity.

FORMULAE

QUANTITY / LAW	FORMULA	SYMBOLS, UNITS & CONDITIONS
Centripetal force	$F = \frac{mv^2}{r} = mr\omega^2 = mv\omega$	F = centripetal force (N); m = mass (kg); v = speed (m s^{-1}); r = radius (m); ω = angular velocity (rad s^{-1})
Horizontal circle on a string / conical pendulum	$T\sin\theta = \frac{mv^2}{r}$; $T\cos\theta = mg$; $\tan\theta = \frac{v^2}{rg}$	T = tension (N); θ = angle of string to the vertical; m = mass (kg); v = speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
Level bend (friction provides F)	$\mu mg \geq \frac{mv^2}{r}$; $v_{max} = \sqrt{\mu gr}$	μ = coefficient of friction; m = mass (kg); v_{max} = maximum speed without skidding (m s^{-1}); r = radius of bend (m)
Banked bend, no friction	$\tan\theta = \frac{v^2}{rg}$	θ = angle of banking; v = speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
Vertical circle, top	$T + mg = \frac{mv^2}{r}$	T = tension (N); m = mass (kg); v = speed at the top (m s^{-1}); r = radius (m)
Vertical circle, bottom	$T - mg = \frac{mv^2}{r}$	T = tension (N); m = mass (kg); v = speed at the bottom (m s^{-1}); r = radius (m)
Minimum speed at top of vertical circle	$v = \sqrt{gr}$	Condition T = 0 at the top; v = minimum speed (m s^{-1}); r = radius (m); g = gravitational acceleration (m s^{-2})
Top of a convex bridge	$mg - N = \frac{mv^2}{r}$	N = normal reaction (N); m = mass (kg); v = speed (m s^{-1}); r = radius of curvature (m); loop or bridge
Bottom of a dip	$N - mg = \frac{mv^2}{r}$	N = normal reaction (N); m = mass (kg); v = speed (m s^{-1}); r = radius of curvature (m); loop or bridge

HOW TO ANSWER (MPM STYLE)

- Draw a free-body diagram, identify which force(s) point towards the centre, and write "resultant force towards the centre = $\frac{mv^2}{r}$".
- Resolve forces vertically (equilibrium, no vertical acceleration) and horizontally (centripetal) for conical pendulum and banked-bend problems.
- For vertical circles state the sign convention: towards the centre positive, so the weight is added at the top and subtracted at the bottom.

- For an explanation of skidding: the required centripetal force $\frac{mv^2}{r}$ exceeds the maximum friction μmg, so the car moves tangentially outward.

SECTION A TRAPS

Trap: Centrifugal force is not an extra force on the body in an inertial frame; do not include $\frac{mv^2}{r}$ as an additional force in the free-body diagram.

Trap: Weight and tension at the top of a vertical circle both act towards the centre, so $T = \frac{mv^2}{r} - mg$; at the bottom $T = \frac{mv^2}{r} + mg$ (tension is largest at the bottom).

Trap: Fixed ω: $F \propto r$. Fixed v: $F \propto \frac{1}{r}$. Mixing these up gives the inverted-ratio option.

Trap: In $v_{max} = \sqrt{\mu g r}$ the mass cancels; a heavier car does not have a larger safe speed.

TRICKS

Trick: Doubling v quadruples F; doubling r at fixed v halves F; at fixed ω doubling r doubles F.

Trick: Centripetal force does no work because $F \perp v$; so the kinetic energy of uniform circular motion is constant.